Training of ML models

- Linear Regression
- Decision Tree Regressor
- Random forest Regressor

Regression → predicting a continuous value.

Tensorflow → Tensorflow is a Opensource library used to build and Train machine learning & neural n/w models.

[Tensorflow is a Tool that helps Computer learn from data & make predictions]

Analogy → Imagine you are a teacher teaching a student.

You give examples → (data)

Student tries to learn → (model training)

Makes mistakes → (error)

You correct → (backpropagation)

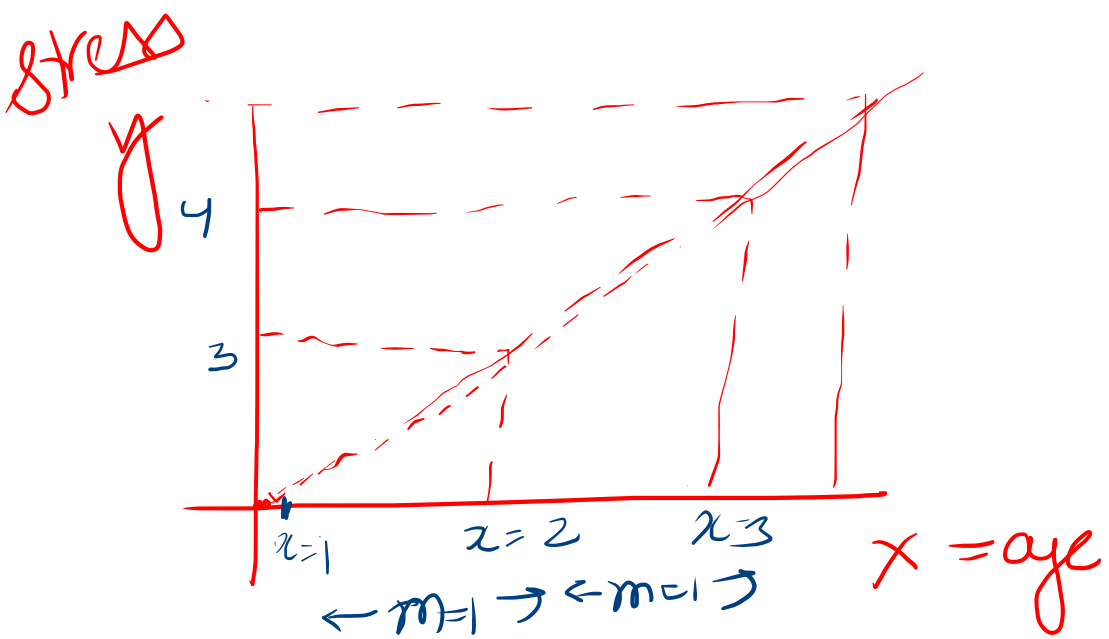
Student improves → (better predictions)

Tensor flow = the class Room + tools that make teaching easy.

How Tensorflow works :-

1. Input data — Images, text, numbers.
2. Model creation → Build neural N/w layers.
3. Training → model learns pattern.
4. Prediction → Gives o/p (e.g spam / notspam)

Linear Regression → Tries to find a straight line relationship b/w input & output



$$y = mx + b$$ formula

b → base value

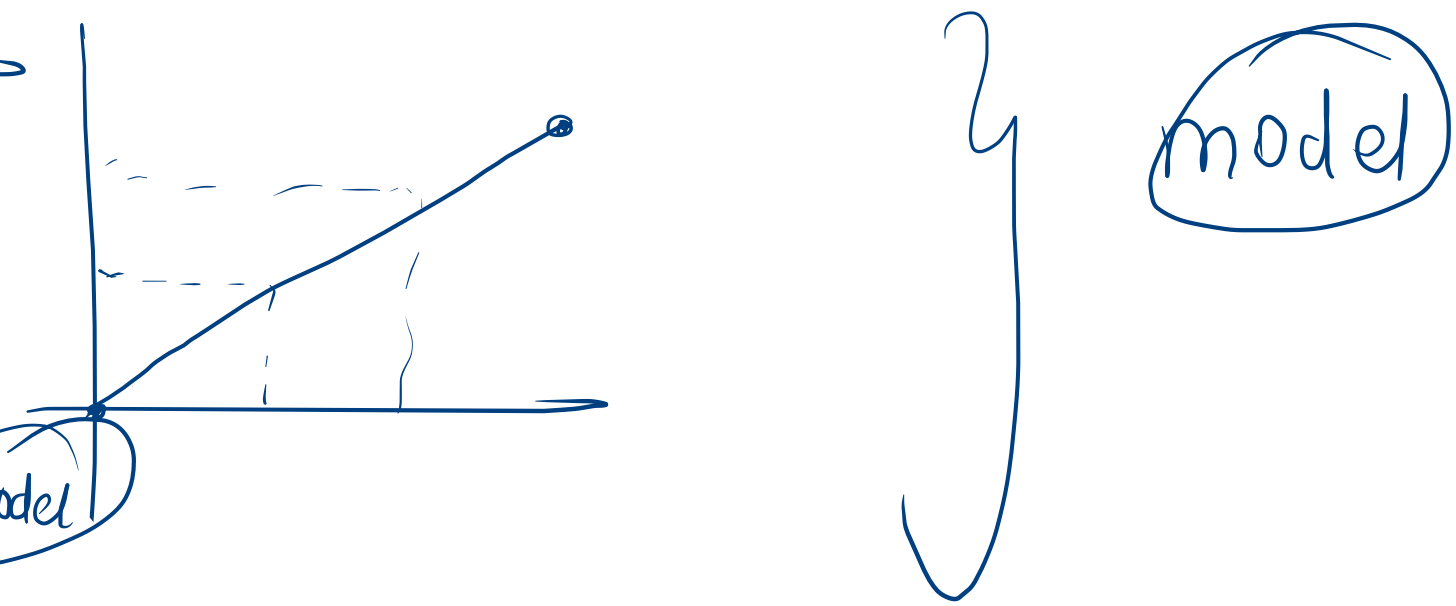
m → how fast x is increasing

Analogy → fixed year increment Job.

Every year → Salary increases by $ 10,000

So, the growth is constant & Predictable.

Experience	Salary
1	$ 3000
2	$ 4000
3	$ 5000
5.5	



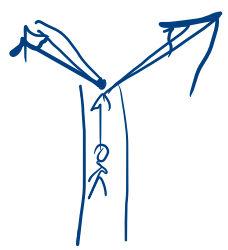
Decision Tree Regression

Split data into rules (if-else conditions)

e.g. Manager deciding salary → If experience < 2 → low salary

If experience b/w 2-4 → medium

If experience > 4 → high



If else Statement

Is experience < 3?

yes → Avg salary = $70000

NO → Is experience < 5?

yes → Avg salary = $90000

NO → Avg Salary = $100000000

Decision Tree → Instead of formulas, it creates rules (if else)

Real world use case → ① loan approval Risk.

→ ② customer segmentation.

③ Medical diagnosis.

Wallmart

④ Pricing based on categories.

* Instead of one decision tree → build many trees & take Average prediction.

Random forest Regressor

* You wanted to buy house :-

You ask 1 property dealer → may be biased

you ask 50 dealers → take avg → more accurate.

that Random forest.

Neural N/w from scratch

1. Write all math (matrix multiplication ...)
2. ~~Implement~~ Correct the model manually
3. Optimize weights step by step.
4. Use CPU / GPU manually

Analogy

w/o Tensorflow → building a car from raw material

with Tensorflow = using ready made engine

↳ Handles complex maths automatically
Perform Training efficient